Submitted By:

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Assignment:

LAB -2

Submitted To:

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For the Fulfillment Of:

FUNDAMENTALS OF PROGRAMMING AND DATA SCIENCE

DEPARTMENT OF COMPUTER ENGINEERING

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Task No.1:

Program for decimal equivalent of binary numbers.

Solution:

```python
# Decimal equivalent of binary number
binary= input("Enter a Binary Number: ")
if all(d in '01'for d in binary):
    decimal=0
    power=0
    for digit in reversed(binary):
        decimal+= int(digit) * (2**power)
        power+=1
        print("Decimal Equivalent is:",decimal)
else:
   print("Invalid binary number !")
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS Code> & C:\U
Enter a Binary Number: 101
Decimal Equivalent is: 1
Decimal Equivalent is: 1
Decimal Equivalent is: 5
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS Code>
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\dell\AppData\Local\Programs\Microsoft VS
Enter a Binary Number: 6
Invalid binary number !
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS
```

Task No.2:

Program for counting vowels and consonants.

Solution:

```python
# Calculating number of vowels and consonants
sentence=input("Enter a sentence: ")
vowels="aeiouAEIOU"
vowel_count=0
consonant_count=0
for char in sentence:
    if char.isalpha():
     if char in vowels:
        vowel_count+=1
     else:
```

```
        consonant_count+=1
print("Number of vowels: ",vowel_count)
print("Number of consonants: ",consonant_count)
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\dell\AppData\Local\Programs\Microsoft VS Code> & C
Enter a sentence: I am a student
Number of vowels:  5
Number of consonants:  6
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS Code> 
```

Task No.3:

Identifying prime numbers by giving a range as input and taking their sum.

Solution:

```
start =int(input("Enter starting number: "))
end=int(input("Enter ending number: "))
prime_sum = 0
print('Prime numbers are: ')
for num in range(start,end + 1):
    if num > 1:
        is_prime=True
        for i in range(2,int(num/2)+1):
            if num % i==0:
                is_prime=False
                break
        if is_prime:
            print(num)
            prime_sum+=num
print("Sum of prime numbers: ",prime_sum)
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\
Enter starting number: 9
Enter ending number: 18
Prime numbers are:
11
13
17
Sum of prime numbers:  41
PS C:\Users\dell\AppData\Local\Programs\
```

Task No.4:

Generating a Random Password.

Solution:

```
#RANDOM PASSWORD
import random
import string
length = int(input("Enter password length: "))
use_upper = input("Include uppercase letters? (y/n): ")
use_lower=input("Include lowercase letters? (y/n): ")
use_digits=input("Include digits? (y/n): ")
use_special=input("Include special characters? (y/n): ")
characters = ""
if use_upper =='y':
    characters += string.ascii_uppercase
if use_lower =='y':
    characters += string.ascii_lowercase
if use_digits =='y':
    characters += string.digits
if use_special =='y':
    characters += string.punctuation
password= ""
for i in range(length):
    password+=random.choice(characters)
print("Generated Password: ",password)
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Mic
Enter password length: 8
Include uppercase letters? (y/n): y
Include lowercase letters? (y/n): y
Include digits? (y/n): y
Include special characters? (y/n): y
Generated Password:  pAjQB~Al
```

<u>Task No.5:</u>

Making a pattern using nested loop.

Solution:

```
rows=5
for i in range(1,rows + 1):
    for j in range(i):
        print("*",end="")
    print()
for i in range(rows -1,0,-1):
    for j in range(i):
        print("*",end="")
    print()
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Microsof
*
**
***
****
*****
****
***
**
*
PS C:\Users\dell\AppData\Local\Programs\Microsof
```